

LAB 3

1. Diagrams:

System Boundary:

- Online submission of borrowing requests
- Approval and scheduling workflow
- Real-time tracking of equipment status
- Automated due date reminders and notifications
- Reporting and analytics tools
- Role-based user access and management

External entities:

- Borrower (Lecturer / Student)
- Approver (Department Head / Lab Technician)
- Inventory Staff
- Admin
- Notification Service (Email/SMS/Push)
- Payment/Finance System (optional, if deposits/fines apply)
- Timer System (due to auto-sending notifications)

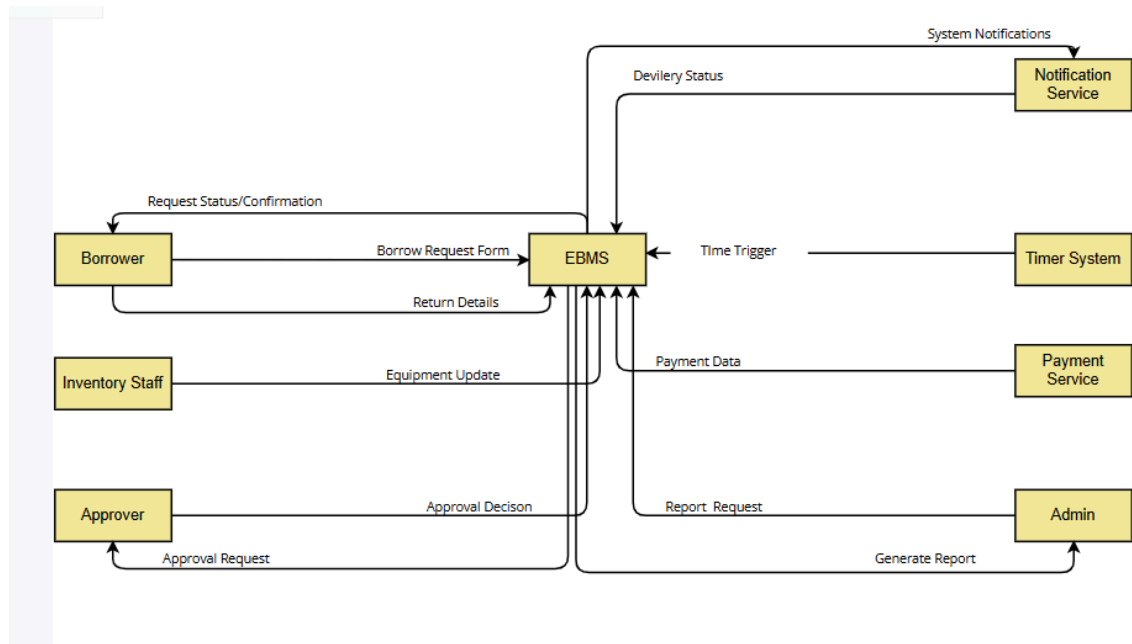
Processes (for level-0 DFD):

- Handle Borrow Request
- Approve / Review Request
- Manage Inventory & Returns
- Notifications & Reminders
- Admin & Reporting

Data Stores (for level-0 DFD):

- Equipment
- Borrow Records
- Users
- Maintenance Records
- Reports
- Notifications Log

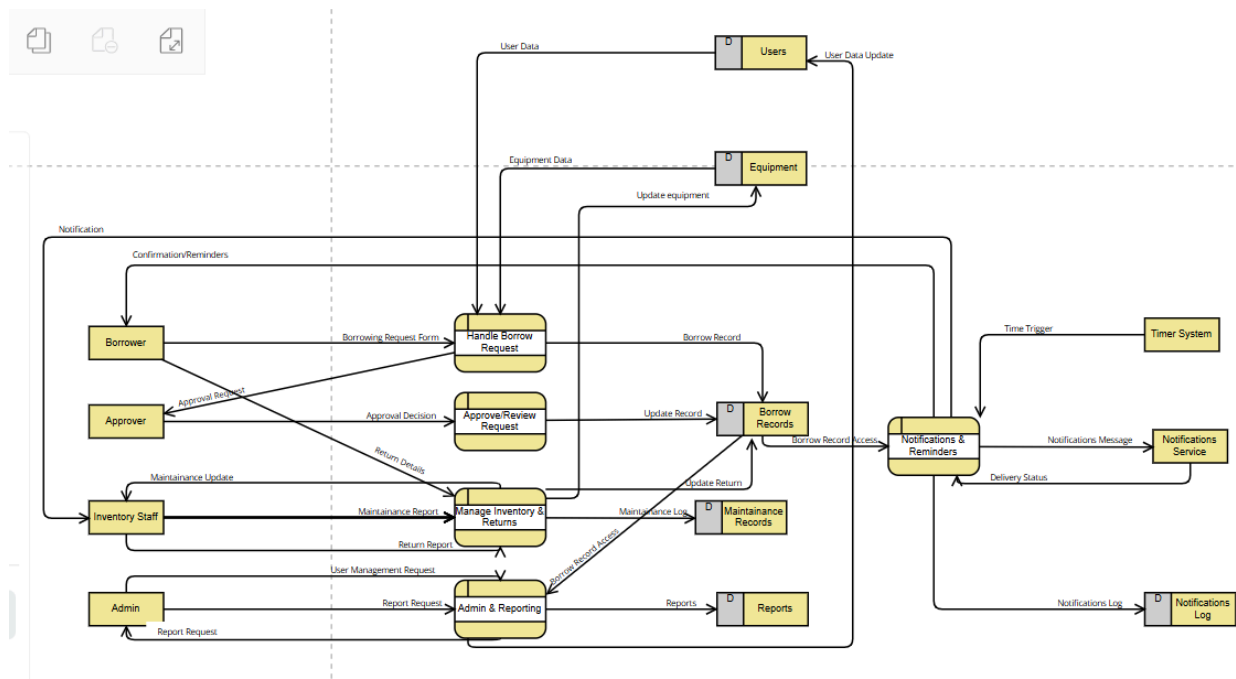
Context Diagram:



In case the words are too blurred, please click the following drive link T-T:

<https://drive.google.com/file/d/1NGjiVaDXVT9YC8Uxq09L7FSVuIB6r3F3/view?usp=sharing>

Level-0 DFD:



In case the words are too blurred, please click the following drive link T-T:

<https://drive.google.com/file/d/1HnNqtxDJvm4kduv00BhcGSAEgQhDfceb/view?usp=sharing>

2. Data Dictionary:

(a) A. Data Elements (DE)

ID	Element Name	Description	Type / Format	Domain / Constraint	Example
DE01	DeviceID	Unique identifier of equipment	String(20)	NOT NULL	EQP-001
DE02	DeviceName	Name of equipment	String(50)	Required	Projector
DE03	UserID	Unique ID of user (lecturer/student)	String(20)	NOT NULL	ST12345
DE04	UserName	Name of borrower	String(50)	Required	Nguyen Van A
DE05	Role	User role (Borrower / Approver / Staff / Admin)	Enum	Predefined list	Borrower
DE06	RequestID	Unique borrow request ID	String(20)	NOT NULL	REQ-2025-001

DE07	BorrowDate	Requested borrow date	Date	≥ today	2025-11-10
DE08	ReturnDate	Planned return date	Date	≥ BorrowDate	2025-11-12
DE09	Quantity	Number of items requested	Integer	≤ available stock	2
DE10	Purpose	Purpose of borrowing	String(100)	Optional	Classroom lecture
DE11	Status	Request status (Pending / Approved / Rejected / Returned / Overdue)	Enum	Predefined list	Approved
DE12	ApprovalDecision	Approver's choice	Enum	Approved / Rejected	Approved
DE13	ApprovalDate	Date of approval	Date	Optional	2025-11-06
DE14	MaintenanceStatus	Current equipment condition	Enum	Available / Under Maintenance	Available

DE1 5	ReportType	Type of report requested	Enum	Borrowing / Late Returns / Utilization	Borrowing
DE1 6	ReportPeriod	Time range of report	String(20)	Optional	Q1-2025
DE1 7	NotificationID	Unique identifier for notification	String(20)	Auto-generated	NTF-102
DE1 8	MessageContent	Content of notification	Text	Required	"Your borrow request has been approved."
DE1 9	DeliveryStatus	Notification delivery result	Enum	Delivered / Failed / Pending	Delivered
DE2 0	Timestamp	Date and time of record or message	DateTime	Auto-generated	2025-11-05 09:30

(b) B. Data Structures (DS)

ID	Structure	Components (DE)	Mandatory	Notes
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DS0 1	BorrowRequest	DE06, DE03, DE07, DE08, DE09, DE10, DE11	Yes	Each borrow request made by a borrower
DS0 2	ApprovalRecord	DE06, DE12, DE13, DE03	Yes	Records approval/rejection of a request
DS0 3	EquipmentRecord	DE01, DE02, DE14	Yes	Equipment catalog and status
DS0 4	UserAccount	DE03, DE04, DE05	Yes	User credentials and roles
DS0 5	MaintenanceReport	DE01, DE14, DE20	Optional	Logs maintenance activities
DS0 6	ReportRequest	DE15, DE16	Optional	Used by Admin to generate reports
DS0 7	NotificationMessage	DE17, DE03, DE18, DE19, DE20	Yes	Message sent to user via Notification Service

(c) C. Data Stores (D)

ID	Name	Description	Key Field	Updated by (Process)
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D 1	Equipment	Contains all equipment details and status	DeviceID	P3 Manage Inventory & Returns
D 2	Borrow Records	Stores all borrow request data	RequestID	P1, P2, P3
D 3	Users	Stores user credentials and roles	UserID	P5 Admin & Reporting
D 4	Maintenance Records	Logs all maintenance updates	DeviceID	P3 Manage Inventory & Returns
D 5	Reports	Stores generated report data	ReportID	P5 Admin & Reporting
D 6	Notifications Log	Logs sent notifications and delivery results	Notification ID	P4 Notifications & Reminders

(d) D. Data Flows (F)

ID	Name	Source	Destination	Structure	Purpose
F0 1	Borrow Request Form	Borrower	P1 Handle Borrow Request	DS01	Submit new borrow request

F0 2	Confirmation / Reminder	P4	Borrower	DS07	Notify borrower of request status or due date
F0 3	Return Details	Borrower	P3 Manage Inventory & Returns	DS01	Record returned items
F0 4	Approval Request	P1	Approver	DS01	Request approval from approver
F0 5	Approval Decision	Approver	P2 Approve / Review Request	DS02	Approver's decision
F0 6	Equipment Update / Maintenance Info	Inventory Staff	P3	DS05	Report equipment status or maintenance
F0 7	System Notifications	P4	Notification Service	DS07	Send notifications externally
F0 8	Delivery Status	Notificatio n Service	P4	DS07	Confirm notification delivery
F0 9	Time Trigger	Timer System	P4	DS07	Start reminder or notification routine

F1 0	Report Request	Admin	P5	DS06	Admin requests reports
F1 1	Generated Report	P5	Admin	DS06	Generated report data
F1 2	Equipment Data	D1	P1	DS03	Get equipment availability
F1 3	Borrow Record	P1/P2/P3	D2	DS01	Store or update borrow record
F1 4	Borrow Record Access	D2	P4/P5	DS01	Retrieve records for reminders/reports
F1 5	Return Update	P3	D1	DS03	Update equipment status after return
F1 6	Maintenance Log	P3	D4	DS05	Save maintenance details
F1 7	Notification Log	P4	D6	DS07	Record all sent notifications
F1 8	User Management Request	Admin	P5	DS04	Manage user accounts

F19	User Data Update	P5	D3	DS04	Save changes to user info
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(a) E. Processes (P)

I D	Process Name	Description	Input s	Output s	Trigger
P1	Handle Borrow Request	Validate and record borrow requests from borrowers	F01, D1, D3	F04, D2	Borrower submits request
P2	Approve / Review Request	Approver reviews and approves/rejects borrow requests	F05, D2	D2, P4	Approver decision
P3	Manage Inventory & Returns	Update equipment status and handle returns	F03, F06, D1	D1, D4, D2	Return submission or maintenance update
P4	Notifications & Reminders	Send notifications and reminders to users	F09, D2, D6	F07, F08, F02	Timer or process events
P5	Admin & Reporting	Generate reports and manage user accounts	F10, D2, D3	F11, D5, D3	Admin action

